CDFD Runtime Paper 08: Proto-Regime, Boundary Conditions, and Theoretical Discovery

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Abstract

Executive synthesis. The Proto-Regime is best understood as a boundary-condition discovery candidate: a high-flow, low-constraint state in which ordinary interpretation of Ψ_s breaks down. Earlier drafts described it too strongly. This upgraded paper frames the Proto-Regime as a reproducible runtime stress test and a theoretical hypothesis, not as proof of faster-than- light travel, pre-Big-Bang physics, or deployable vacuum engineering.

Runtime evidence path
Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Definition

The operating ratio is

$$\Psi_s = \frac{\Phi}{C} S M_s. \tag{1}$$

The Proto-Regime refers to simulations where Φ is forced high while C is forced toward a numerical floor. In this limit, ordinary flow-through- constraint interpretation becomes unstable because the denominator approaches zero.

3 Why It Matters

A good runtime must know where its own equations stop behaving like ordinary domain models. The Proto-Regime is one such boundary. It marks a region where the model triggers warnings, finite-value checks, and explicit caveats.

4 Experiment Surface

The local script `experiments/discover_proto_regime.py` is the named entrypoint for probing this boundary. Related discovery machinery appears in:

- `experiments/notebooks/discovery_frontier_sweep.py`;
- `experiments/reports/NEW_DISCOVERIES_REPORT.md`;
- `experiments/reports/FRONTIER_DISCOVERIES.md`.

These reports are simulation logs. A high novelty score or new schema name means the engine found an unusual modeled state, not that nature has accepted the interpretation.

5 Mathematical Boundary

Let $C = \epsilon$ with $\epsilon > 0$ small. Then

$$\Psi_s \sim \frac{\Phi}{\epsilon} S M_s. \tag{2}$$

For fixed Φ, S, and M_s, Ψ_s diverges as $\epsilon \to 0$. The question is not whether the number becomes large; that is expected. The real question is whether the state produces reproducible structure under bounded regularization, or whether it is merely a division-by-small-number artifact.

6 Part I Boundary Marker

The active Part I boundary marker used here is the Paper XII superconductivity reallocation sample with transport index

$$\mathcal{S}_{\mathrm{SC}} = 4.187183192107868$$

at the row where $T/T_c = 0.5287891440501045$. That row appears in the Part I Paper XII superconductivity reallocation table and belongs to the Part I transport-test program [5]. The runtime Proto-Regime is therefore written as a boundary family anchored to the Part-I-derived 4.187183192107868 transport marker.

7 Interpretation Rules

The Proto-Regime may be used as:

- a numerical stress test;
- a boundary marker for model validity;
- a candidate theoretical state for future mathematical analysis;
- a warning system for non-physical parameter choices.

It does not establish:

- frictionless travel;
- physical vacuum manipulation;
- a device claim;
- a clinical or engineering recommendation.

8 Falsification and Controls

Controls include:

1. increasing the constraint floor ϵ and checking whether the pattern vanishes;
2. adding finite S and M_s caps;
3. varying grid resolution and time step;
4. rerunning with different seeds and boundary conditions;
5. comparing against a known numerical singularity test.

9 Connection to the CLI

The CLI now needs a safe boundary-test command. Until then, Proto-Regime experiments remain Python scripts and report entries. Any future CDFL version needs an explicit warning when C is below a declared physical or domain-specific floor.

10 Conclusion

The Proto-Regime is important precisely because it is dangerous to overread. It marks a boundary where the runtime can teach its authors which claims need more mathematics, more controls, and stronger validation before they become physics.

References

[1] Steve Bico Mujjabi, MD. *Vortex Stability in a Constrained Transport Vacuum and the Origin of the Fine-Structure Constant.* CDFD Part I Paper I, preprint, 2026.

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[6] Steve Bico Mujjabi, MD. *CDFD Part II: Origins of Life and Tri-Regime Bioenergetics.* Public release archive, 2026, `https://doi.org/10.5281/zenodo.20250821`.